

DES Decryption Using Reverse Key Order

17. Write a C program for DES algorithm for decryption, the 16 keys (K1, K2, ..., K16) are used in reverse order. Design a key-generation scheme with the appropriate shift schedule for the decryption process.

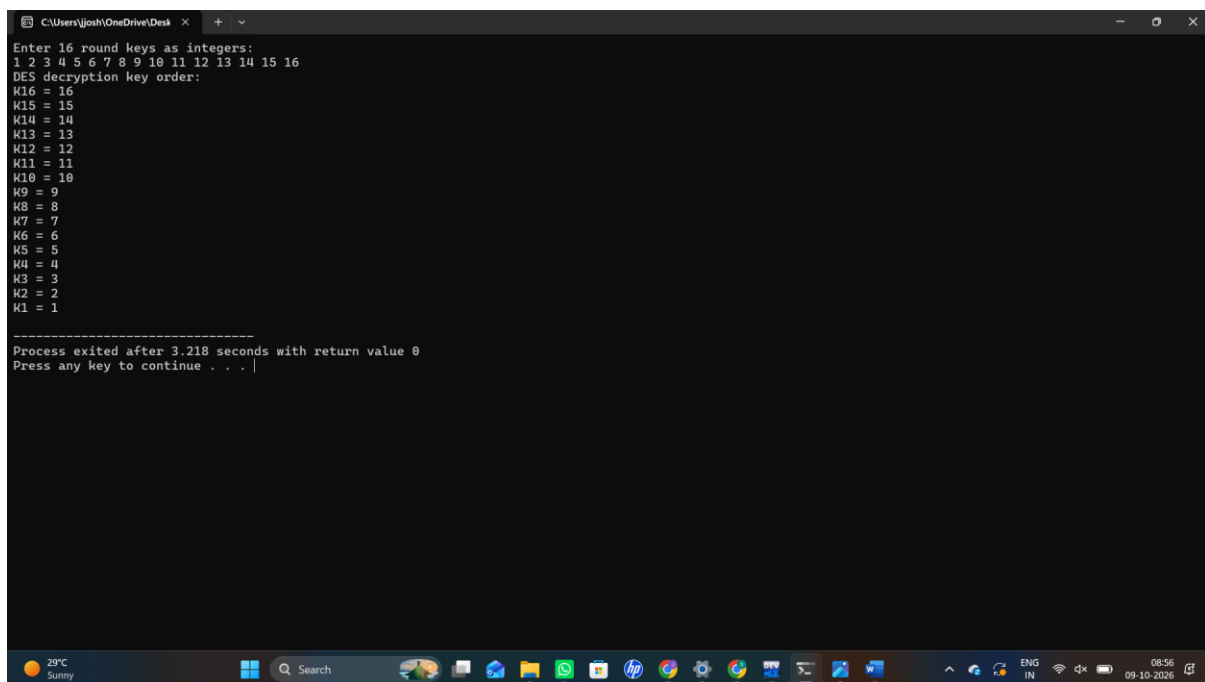
Aim:

To write a C program demonstrating that DES decryption uses the 16 round keys in reverse order.

Algorithm:

1. Start the program.
2. Read the 16 encryption round keys.
3. Reverse the order of the keys.
4. Display the round keys for decryption.
5. Stop.

Program :



```
C:\Users\josh\OneDrive\Desktop >
Enter 16 round keys as integers:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
DES decryption key order:
K16 = 16
K15 = 15
K14 = 14
K13 = 13
K12 = 12
K11 = 11
K10 = 10
K9 = 9
K8 = 8
K7 = 7
K6 = 6
K5 = 5
K4 = 4
K3 = 3
K2 = 2
K1 = 1

-----
Process exited after 3.218 seconds with return value 0
Press any key to continue . . .
```

Input/Output:

```
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Enter 16 round keys as integers:
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16
DES decryption key order:
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-----
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```

Result:

The program demonstrates reverse ordering of the 16 DES round keys for decryption.